GHISLAIN FRANK TONTSA APALA

EMBEDDED SOFTWARE ENGINEER

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Embedded software engineer with hands-on experience in C firmware, RTOS, Embedded Linux. Real-world experience in automotive & IoT. Actively seeking a first full-time role in embedded software engineering across Europe.

Work experience

Embedded Software Developer Intern (6 months)

From March 2025 to August 2025 EFI Automotive Lyon, France

- Implemented a UDS server (ISO 14229) over CAN on a dsPIC33CK256MP508 microcontroller running FreeRTOS.
- Improved memory management and FreeRTOS task scheduling to reduce power consumption and enhance system robustness.
- Integrated a watchdog mechanism to prevent system lock-ups and ensure safe software execution.
- Conducted functional testing with Vector CANoe to validate compliance with ISO standard
- V-cycle development.

Reference: remi.vaganay@efiautomotive

Freelance IoT Developer (6 months)

From August 2024 to November 2024 OXYNODE Belfort

- Developed JavaScript decoders to process and normalize IoT sensor data transmitted over the LoRaWAN network on the Live Objects platform.
- Implemented unit tests and validated received frames, ensuring reliability and consistency of the processed data.

Embedded Software Developer Intern (6 months)

From September 2023 to February 2024 INMAN Haguenau

- Developed Firmware for an electronic water mixer using proportional valves controlled by a PIC18F16K40 microcontroller.
- Developed an embedded tool based on ESP32 to test and operate the InSens connected shower's hydraulic system via bidirectional RS232 communication.

Education

Computer Engineering :Embedded systems and Robotics

From September 2022 to 2025 Université Technologique de Belfort Montbéliard Belfort

Computer Engineering (ERASMUS) :Embedded Systems and Robotics

From October 2024 to March 2025 RPTU Kaiserslautern Kaiserslautern, RP, Germany

Bachelor's Degree in Computer Science : Embedded and Mobile systems

From September 2019 to June 2022 ISET Nabeul,Tunisia

Projects

- **Real-Time Embedded System with Zephyr RTOS — ARM Cortex-M, C**
  Multitask application development : thread management, timers, GPIO via Device Tree, West build pipeline.

- **Custom Embedded Linux Distribution — Yocto Project, Raspberry Pi**
  Full Linux build from scratch : U-Boot configuration, custom kernel compilation, Device Tree customization, Yocto layers & recipes.

DISTINCTIONS

- **Eiffel Excellence Scholarship Laureate** *(2022–2025)*

- **Tunisia–Cameroon Cooperation Scholarship Laureate** *(2019–2022)*

Languages

French
Native language

English
Native language

My Portfolio

Click here

Social networks

@FrankApala
Frank Tontsa Apala

Technical skills

Programming Languages:
C, C++, Python, Java

Operating Systems:
Embedded Linux , Zephyr RTOS, FreeRTOS

Hardware Architectures:
ARM Cortex-A, ARM Cortex-M, PIC 8/16/32-bit

Microcontrollers & Platforms:
STM32, ESP32, Raspberry Pi, PIC/dsPIC

Communication Protocols:
CAN, UART, SPI, I^2C, MQTT

Build Systems and Automation:
Makefile, CMake, Yocto

Tools & Debugging:
Git, GDB, MPLAB ICD, Oscilloscope, CANoe

Containerization & Virtualization
Docker, QEMU

Interests

Technical content creation
(TA Embedded)

Sports (Gym Workout)